

This PDF originates from <https://sheafofthoughts.org>.

Why Fermat had (probably) no proof?

2025-12-18

Introduction

According to Dickinson (1938), Fermat stated the following in 1637:

It is impossible to separate a cube into two cubes, or a biquadrate into two biquadrates, or in general any power higher than the second into the power of like degree; I have discovered a truly remarkable proof which this margin is too small to contain.”

This theorem is now known as Fermat’s Last Theorem, and it can be stated as follows.

Theorem 0.1 (Fermat’s last theorem). *Let $n \geq 3$ be an integer. The equation*

$$x^n + y^n = z^n$$

has no non-trivial solutions $x, y, z \in \mathbb{Z}$.

First, observe that we can reduce to consider only triples $(x, y, z) \in \mathbb{Z}^3$ with $\gcd(x, y, z) = 1$. In particular, in this situation x, y, z are pairwise coprime: for instance, if a prime ℓ divides x and z , then it divides $z^q - x^q = y^q$, hence it divides y , contradicting $\gcd(x, y, z) = 1$. Moreover, Fermat himself proved the theorem in the case $n = 4$; therefore, using basic properties of powers, it is enough to prove the theorem for $n = q$, where q is an odd prime.

To prove Theorem 0.1, we could naively proceed in the following way. Fix an odd prime $q \geq 3$ and suppose that we had a non-trivial solution $x, y, z \in \mathbb{Z}$ of

$$x^q + y^q = z^q \tag{1}$$

Then, reducing this equation modulo every prime p such that $p \nmid xyz$, we get a non-trivial solution modulo p . Therefore, only finitely many primes p can fail to admit a non-trivial solution modulo p . If we proved that for infinitely many primes p the reduction of Equation 1

modulo p has no non-trivial solution, then we would conclude that Equation 1 has no integer solutions at all.

We will prove that this method cannot work. Indeed, we have the following.

Theorem 0.2. *There exists an integer $N \geq q$ such that for every prime $p > N$, Equation 1 has a non-trivial solution in $\mathbb{F}_p$.*

Therefore, there are only finitely many primes with no non-trivial solution, which proves that the method cannot work.

Warm up: a special case

Proposition 0.1. *Fix a prime q . There are infinitely many primes p such that 2 has a q -th root in $\mathbb{F}_p$.*

From this proposition we obtain that the equation $x^q + y^q = z^q$ has a non-trivial solution modulo p for infinitely many primes p . Indeed, if $z^q = 2$ in $\mathbb{F}_p$, we can choose $x = y = 1$ and get

$$1^q + 1^q = z^q.$$

of Proposition 0.1. This proposition is an easy corollary of Corollary 5 you can find in a [previous post](#) about the Chebotarev density theorem, applied to the algebra $A := \mathbb{Z}[X]/(X^q - 2)$. $\square$

Remark 0.1. Observe that the Chebotarev density theorem only tells us that we find infinitely many primes giving a solution to the equation, but it also tells us that the set of primes we find has density less than 1. Therefore, it is not sufficient to establish what we want.

Proof of Theorem 0.2

The idea to prove Theorem 0.2 is to use a geometric fact. In particular, we want to study the projective curve

$$X := V_+(X^q + Y^q - Z^q) \subseteq \mathbb{P}_{\mathbb{F}_p}^2,$$

where $p \neq q$ are primes. Finding a non-trivial solution in $\mathbb{F}_p$ is the same as finding a point $[x : y : z] \in X(\mathbb{F}_p)$ such that $xyz \neq 0$.

After proving that X satisfies the hypotheses of the [Weil conjectures](#), we will use them to establish, for p large enough, a bound

$$\#X(\mathbb{F}_p) > 3q,$$

where $X(\mathbb{F}_p)$ denotes the $\mathbb{F}_p$ -rational points. We will also prove that the points where one coordinate vanishes are $\leq 3q$. Therefore, we will get the theorem as a direct consequence of the bound.

We now state the Weil conjectures in the form we need.

Theorem 0.1 (Special case of the Weil conjectures). *Suppose that X is a smooth, geometrically irreducible, projective curve over the field $\mathbb{F}_p$, and let g be the genus of X . The zeta function $\zeta(X, t)$ of X is, by definition,*

$$\zeta(X, t) := \exp \left(\sum_{m=1}^{\infty} \#X(\mathbb{F}_{p^m}) \frac{t^m}{m} \right).$$

Then $\zeta(X, t)$ is a rational function and can be written as

$$\zeta(X, t) = \frac{P(t)}{(1-t)(1-pt)},$$

where $P(t) \in \mathbb{Z}[t]$ has degree $2g$ and factors as

$$P(t) = \prod_{i=1}^{2g} (1 - \alpha_i t) \quad \text{with} \quad |\alpha_i| = \sqrt{p}.$$

Now we are ready to prove [Theorem 0.2](#).

of [Theorem 0.2](#).

- **Step 1:** *There are at most $3q$ points of $X = V_+(X^q + Y^q - Z^q) \subseteq \mathbb{P}_{\mathbb{F}_p}^2$ such that at least one of the coordinates x, y, z vanishes.*

Proof: Consider the line $z = 0$. On this line, the equation becomes $X^q + Y^q = 0$, which is a homogeneous equation of degree q on $\mathbb{P}_{\mathbb{F}_p}^1$, hence it has at most q solutions over any field. Repeating the same argument for the lines $x = 0$ and $y = 0$ gives the claim.

- **Step 2:** *X is a smooth, geometrically irreducible, projective curve of genus $g = \frac{(q-1)(q-2)}{2}$.*

Proof: X is projective by construction. It is smooth because $p \neq q$: if $F := X^q + Y^q - Z^q$, then

$$\frac{\partial F}{\partial X} = qX^{q-1}, \quad \frac{\partial F}{\partial Y} = qY^{q-1}, \quad \frac{\partial F}{\partial Z} = -qZ^{q-1},$$

and since $q \not\equiv 0 \pmod{p}$, these three partial derivatives cannot vanish simultaneously at any closed point of $\mathbb{P}_{\mathbb{F}_p}^2$.

We now prove that X is geometrically irreducible. This means that

$$X \times_{\mathrm{Spec}(\mathbb{F}_p)} \mathrm{Spec}(\overline{\mathbb{F}_p})$$

(where $\overline{\mathbb{F}_p}$ is an algebraic closure) is an irreducible topological space. If it were reducible, then F would factor in $\overline{\mathbb{F}_p}[X, Y, Z]$ as a product $F = GH$ with $\deg(G), \deg(H) > 0$. By Bézout's theorem we get that $V_+(G) \cap V_+(H) \neq \emptyset$. Pick $P \in V_+(G) \cap V_+(H)$. Then, for every $W \in \{X, Y, Z\}$,

$$\frac{\partial F}{\partial W} = G \frac{\partial H}{\partial W} + H \frac{\partial G}{\partial W},$$

so all partial derivatives of F vanish at P , contradicting smoothness.

Finally, since X is a smooth plane curve of degree q , its genus is $g = \frac{(q-1)(q-2)}{2}$ (by [genus-degree formula](#)).

- **Step 3:** Apply Theorem [0.1](#).

From (in the following we work with formal series $\mathbb{C}[[t]]$)

$$\zeta(X, t) = \exp \left(\sum_{m=1}^{\infty} \#X(\mathbb{F}_{p^m}) \frac{t^m}{m} \right)$$

we get

$$\#X(\mathbb{F}_p) = \left(\frac{d}{dt} \log(\zeta(X, t)) \right) \Big|_{t=0}.$$

Using

$$\zeta(X, t) = \frac{P(t)}{(1-t)(1-pt)} \quad \text{and} \quad P(t) = \prod_{i=1}^{2g} (1 - \alpha_i t),$$

we obtain

$$\#X(\mathbb{F}_p) = \left(\frac{d}{dt} (\log(P(t)) - \log(1-t) - \log(1-pt)) \right) \Big|_{t=0}$$

hence

$$\#X(\mathbb{F}_p) = p + 1 - \sum_{i=1}^{2g} \alpha_i.$$

Since $|\alpha_i| = \sqrt{p}$ for all i , we deduce the bound

$$\#X(\mathbb{F}_p) \geq p + 1 - 2g\sqrt{p} = p + 1 - (q-1)(q-2)\sqrt{p}.$$

Hence, for p large enough, we have $\#X(\mathbb{F}_p) > 3q$ (indeed the right-hand side approaches $+\infty$ when p approaches $+\infty$). By Step 1, this implies that there must be a point $[x : y : z] \in X(\mathbb{F}_p)$ with $xyz \neq 0$, i.e. a non-trivial solution modulo p .

□

Extending to p -adic solutions

We can extend the theorem thanks to [Hensel's lemma](#). In particular, we can say the following.

Theorem 0.2 (Extension of Theorem 0.2). *There exists an integer $N \geq q$ such that for every prime $p > N$, the Fermat equation (Equation 1) has a non-trivial solution in $(\mathbb{Z}_p)^\times$.*

From Theorem 0.2 we get, in particular, a non-trivial solution in every $\mathbb{Z}/p^r\mathbb{Z}$ for every $r \geq 1$ and for every prime $p > N$.

Since we can factorize every integer as a product of primes, this tells us that if m is an integer having at least one prime divisor $> N$, then $x^q + y^q = z^q$ has a non-trivial solution in $\mathbb{Z}/m\mathbb{Z}$.

of Theorem 0.2. Fix a prime $p > N$. From Theorem 0.2 we know that there is a solution $(\bar{a}, \bar{b}, \bar{c}) \in (\mathbb{F}_p^\times)^3$ such that

$$\bar{a}^q + \bar{b}^q = \bar{c}^q \quad \text{in } \mathbb{F}_p.$$

Choose lifts $a, b, c_0 \in \mathbb{Z}_p^\times$ such that $a \equiv \bar{a}, b \equiv \bar{b}, c_0 \equiv \bar{c} \pmod{p}$. Define the polynomial $g(T) := a^q + b^q - T^q \in \mathbb{Z}_p[T]$.

Then

- $g(c_0) \equiv 0 \pmod{p}$;
- $g'(T) = -qT^{q-1}$, so $g'(c_0) \equiv -q\bar{c}^{q-1} \not\equiv 0 \pmod{p}$ (since $p \neq q$ and $\bar{c} \neq 0$).

By Hensel's lemma there is a unique $c \in \mathbb{Z}_p$ such that $c \equiv c_0 \pmod{p}$ and such that c is a root of g in $\mathbb{Z}_p$. In particular,

$$a^q + b^q = c^q \quad \text{in } \mathbb{Z}_p,$$

so $(a, b, c) \in (\mathbb{Z}_p^\times)^3$ is a non-trivial p -adic solution. □

What do we bring with us

There are some general facts we used in our reasoning that could be used in other contexts. I would like to point out some of these things.

The first one is from Step 2 of the proof of Theorem 0.2.

Proposition 0.1. *Let k be a field and $F(X_1, \dots, X_n) \in k[X_1, \dots, X_n]$, with $n \geq 3$, a homogeneous polynomial. Define the ideal $J := (\frac{\partial F}{\partial X_1}, \dots, \frac{\partial F}{\partial X_n})$. If the intersection between the zeros of J (closed points of $V(J)$) and the roots of F is just $(0, \dots, 0)$ in $\bar{k}^n$, then F is irreducible in $\bar{k}$.*

Similarly, observe that the bound

$$\#X(\mathbb{F}_q) \geq p + 1 - (q - 1)(q - 2)\sqrt{p}$$

deduced in Step 3 of the proof of Theorem 0.2 is true for every smooth, geometrically irreducible, projective curve of genus $\frac{(q-1)(q-2)}{2}$.

Moreover, you can check that the proof of Theorem 0.2 is actually a proof of the following more general fact:

Theorem 0.1. *Let $F(X, Y, Z) \in \mathbb{Z}[X, Y, Z]$ be a homogeneous polynomial, and $J := (\frac{\partial F}{\partial X}, \frac{\partial F}{\partial Y}, \frac{\partial F}{\partial Z})$. Then, if eventually for every p prime the intersection between the zeros of J and the roots of F is $(0, \dots, 0)$ (or empty) in $\overline{\mathbb{F}}_p^3$, F has eventually non-trivial solutions modulo every p .*

Finally, we also proved in Theorem 0.2 the following.

Proposition 0.2. *Let A be a henselian local ring (a local ring where the one-variable Hensel lemma is true) with maximal ideal $\mathfrak{m}$, let k be its residue field, let $f \in A[X_1, \dots, X_n]$, and let $a = (a_1, \dots, a_n)$ be a solution of f in k such that $\nabla f(a_1, \dots, a_n) \neq 0$ in k . Then there exists a solution $(s_1, \dots, s_n)$ of f in A such that $s_i \equiv a_i \pmod{\mathfrak{m}}$.*

Conclusion

The key point is that reducing Fermat's equation modulo primes is simply too weak to rule out integer solutions. In fact, for a fixed odd prime exponent q , the Fermat curve

$$X : X^q + Y^q = Z^q$$

has $\mathbb{F}_p$ -points with all coordinates non-zero for every sufficiently large prime p .

Moreover, Hensel's lemma shows that these solutions eventually lift to $\mathbb{Z}_p$, and hence give solutions modulo p^r for all $r \geq 1$.

This helps explain why Fermat's marginal claim is implausible: any proof of Fermat's Last Theorem must use information that cannot be recovered by congruences modulo primes, modulo n , or elementary p -adic methods. However, we don't know whether Fermat could have proven the theorem using other techniques, as he did in the case $n = 4$.

I would like to thank my friend who suggested this topic to me and helped me write this post.

Dickinson, L. J. 1938. *History of the Theory of Numbers*. New York: Chelsea Publishing Company.